

DIVYA JYOTY

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EDUCATION

VIT-AP University, Amaravati, Andhra Pradesh

2023 – 2027

B.Tech, Computer Science and Engineering | CGPA: 9.39/10 | **Expected Graduation:** May 2027

Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Machine Learning, Computer Organization & Architecture, Mathematics, Statistics

Awards: **1st Place**, Artificial Intelligence and Machine Learning Hackathon 2025), IIT-Hyderabad | **1st Place**, Generative AI Hackathon 2025, IIT-Hyderabad | **3rd Place**, Mole and the Byte Hackathon held by ACS Chapter

EXPERIENCE

AI Engineer Intern, IBM–Adroit Technologies — Remote, Bengaluru, Karnataka

May 2026 – July 2026

- Forged personalized learning roadmaps by employing **Python, Flask, and Google Gemini API**, driving end-to-end **API development** and boosting user engagement by **30%** within the generator's initial deployment.
- Designed a **3-tier AI application** architecture spanning frontend, **Flask backend, and Gemini-based AI processing**, driving API development by building **2 RESTful Flask endpoints** as Web Services for user interaction and learning-path generation.
- Formulated and tested advanced prompt templates for **enterprise software** automation of the learning-path business process, **improving AI output** relevance and consistency by **40%** through iterative response refinement.

PROJECTS

GPU Inference Benchmarking | PyTorch · CUDA · FastAPI · TensorRT · Python · Streamlit | [GitHub](#)

Jul 2026 – Aug 2026

- Architected a **deep learning** benchmarking system evaluating model execution across **3 inference configurations: CPU, PyTorch CUDA, and TensorRT**, optimizing algorithms and systems software for low-level hardware performance.
- Architected and optimized custom **CUDA** preprocessing kernels, **decreasing memory bottlenecks and accelerating real-time inference** throughput by **35%** through rigorous testing.
- Optimized GPU hardware utilization by **18%** via **cuDNN** profiling, presenting performance telemetry and visualization through a real-time Streamlit dashboard.

Selective Intelligence | Python · PyTorch · Streamlit Learning · NumPy | [GitHub](#)

Oct 2025 – Nov 2025

- Built a custom **Reinforcement Learning environment** with **1,152 discrete states** to optimize **4 educational intervention strategies** across student proficiency profiles.
- Implemented and benchmarked **Tabular Q-Learning and Deep Q-Networks (DQN)** using PyTorch, a **5,000-capacity experience replay buffer**, and **ε-greedy/Softmax** exploration across **4,000+ simulation episodes**.
- Constructed a real-time Streamlit dashboard to visualize cognitive states, model decisions, and action distributions, enhancing analytic clarity by **25%**.

SkillSwap Hub | React.js · Node.js · Express.js · MongoDB · Socket.IO · Razorpay · JWT | [GitHub](#)

Aug 2025 – Sep 2025

- Initiated a full stack development project using **React.js, Node.js, and MongoDB**, translating business requirements into a **7-collection database schema** with **SQL** reporting queries that enhanced skill discovery and secure user exchanges.
- Designed a real-time messaging solution using **JavaScript** and **WebSockets**, implementing persistent chat history and live typing indicators, reducing message latency by **50%**, with a focus on UI/UX principles.
- Integrated **Razorpay payment processing** with server-side **HMAC-SHA256 signature verification**, JWT authentication, premium subscriptions, and payment audit tracking.

LEADERSHIP

- Former Vice President, IETE Chapter, VIT-AP** — Documented 6+ club initiatives serving 100+ active members, standardizing volunteer onboarding.
- Former Joint Secretary, IETE Chapter, VIT-AP** — Judged 40+ projects across technical and design tracks

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, MATLAB

Systems & Infrastructure: Linux, Bash/Shell Scripting, Docker, Kubernetes, RESTful APIs, SOAP Web Services, Microservices, Cloud concepts

Databases & Cloud: MongoDB, Mongoose, MySQL

Developer Tools: Git, GitHub Actions, CI/CD, pytest, integration testing, code review, debugging, GitHub copilot, DevOps

WebDevelopment: React.js, Node.js, Express.js, Flask, Spring Boot, Socket.IO, WebSockets, JWT,

Machine Learning: Deep Learning, Reinforcement Learning, PyTorch, Scikit-learn, NumPy, Pandas

Web Server: Apache

Certifications: AWS Academy Cloud Foundations, AWS Academy Cloud Architecture (2025)